

GPP

GPP is an R package (version $\geq 4.3.3$) for genomic data analysis and visualization. It is designed for applications in population genetics, genome-wide association studies (GWAS), linkage disequilibrium (LD) analysis, and functional annotation. The package integrates multiple genomic data formats and outputs from various analytical modules, establishing a unified workflow for analysis and visualization, thereby enabling efficient generation of publication-quality scientific figures.

1. System Requirements:

R: 4.3.3 or higher

Operating System: Windows / Linux / macOS

2. Installation

Clone the repository:

```
git clone https://github.com/fu-yujiao/GPP.git
```

Open R and set working directory:

```
setwd("your_directory/GPP")
```

Load GPP:

```
source("GPP_functions.R")
```

3. Supported Input Formats

GPP supports multiple genomic analysis outputs and general tabular files. All these files must be in the same directory as your data:

GAPIT results (GAPIT.FarmCPU.EarHT.csv、 GAPIT.BLINK.R_H.csv、
GAPIT.GLM.G_H.csv、 GAPIT.FarmCPU.B_H.csv、 GAPIT.FarmCPU.R_R.csv、
GAPIT.FarmCPU.G_R.csv、 GAPIT.FarmCPU.B_R.csv)

PLINK results (PLINK.Linear.EarHT.linear)

rMVP results (RMVP.FarmCPU.EarHT.csv)

Genotype files (mdp_SNP_information.txt、 GAPIT.Genotype.Frequency_MAF.csv、 GAPIT.Genotype.View")

PCA files (file.csv、GAPIT.Genotype.PCA.csv、GAPIT.Genotype.PCA_eigenvalues.csv)

Annotation files (Zea_mays.Zm-B73-REFERENCE-NAM-5.0.62.gtf、Bos_grunniens.LU_Bosgru_v3.0.99.gtf)

LD decay results (hb.Out.LDdecay.stat、hlj.Out.LDdecay.stat、qy.Out.LDdecay.stat and yak.Out.LDdecay.stat)

4. Usage

Run GPP in R:

```
source("GPP_functions.R")
```

Example 1: Genotype.View

Required files: Genotype files (mdp_SNP_information.txt, GAPIT.Genotype.Frequency_MAF.csv, GAPIT.Genotype.View).

```
setwd("your_directory/GPP")
```

```
GPP_result = GPP(file_path = "C:\\GAPIT", graph = "genotype")
```

Supported Genotype Features (Individual heterozygosity, marker heterozygosity, minor allele frequency (MAF), genotypic correlation (r), physical distance between neighboring SNPs, and LD decay based on r^2 .)

Example 2: PCA Analysis

Required files: PCA files (file.csv、GAPIT.Genotype.PCA.csv、GAPIT.Genotype.PCA_eigenvalues.csv)

```
setwd("your_directory/GPP")
```

```
GPP(file_path = "C:/GAPIT", graph = "PCA", PCA.pch.2D = 20,
```

```
PCA.cex.2D = 1.1, PCA.alpha = 0.75, PCA.legend.cex = 1.2,
```

```
PCA.legend.outside.3D = TRUE, PCA.legend.inset.3D = c(-0.3, 0),
```

```
PCA.mar.right.3D = 14, PCA.dynamic3D.backend = "plotly",
```

```
PCA.dynamic3D.title = "Interactive 3D PCA", PCA.dynamic3D.marker.size = 4,
```

```
PCA.dynamic3D.opacity = 0.8)
```

Supported PCA Features (2D PCA、3D PCA、 Interactive PCA 、 Multi-group coloring)

Example 3: Circle Manhattan

Required files (GAPIT.Association.GWAS_Results.BLINK.dpoll、
 GAPIT.Association.GWAS_Results.BLINK.EarHT、
 GAPIT.Association.GWAS_Results.FarmCPU.dpoll、
 GAPIT.Association.GWAS_Results.FarmCPU.EarHT、
 GAPIT.Association.GWAS_Results.BLINK.B_R、
 GAPIT.Association.GWAS_Results.BLINK.R_R、
 GAPIT.Association.GWAS_Results.FarmCPU.B_R、
 GAPIT.Association.GWAS_Results.FarmCPU.R_R、
 GAPIT.Association.GWAS_Results.MLM.B_R、
 GAPIT.Association.GWAS_Results.MLM.B_R)

```

model_store <- c("BLINK", "FarmCPU")
Y.names <- c("dpoll", "EarHT")
#model_store <- c("BLINK","FarmCPU","MLM")
#Y.names <- c("R_R", "B_R")

GPP.Circle.Manhattan.Plot(model_store = model_store, Y.names = Y.names,
  cutOff = 0.05, byTraits = FALSE)

```

Supports drawing Circle Manhattan

Example 4: Multi-model Manhattan graph with gene annotations

Required files (GAPIT.BLINK.StandCount.csv、PLINK.linear.StandCount.linear、
 rMVP.FarmCPU.StandCount.csv)

```

GPP.GWAS.Manhattan.Horizontal.MultiSoftware(
  files = c(
    "GAPIT.BLINK.StandCount.csv",
    "PLINK.linear.StandCount.linear",
    "rMVP.FarmCPU.StandCount.csv"),
  out = "Y1.GTF.pdf",
  preset = "pub_large", plot_mode = "stacked",
  annot = "Zea_mays.Zm-B73-REFERENCE-NAM-5.0.62.gtf",
  window_kb = 10,
  out_sig = "Y1.Significant.csv")

```

Example 5: LD plots

Required files: LD decay results (hb.Out.LDdecay.stat、hlj.Out.LDdecay.stat、qy.Out.LDdecay.stat and yak.Out.LDdecay.stat)

```
source("C:/GPP/GPP_functions.R")
setwd("C:/GPP")
files <- c("hb.Out.LDdecay.stat",
          "hlj.Out.LDdecay.stat",
          "qy.Out.LDdecay.stat")
LDplot(files, out = "hb_hlj_qy.LDdecay.pdf")
```

5. File Descriptions

(1) Core Scripts

GPP_functions.R: Main source file containing all GPP functions for genomic analysis and visualization.

(2) Supported Input Files

GAPIT results: GWAS results generated by GAPIT (GLM, MLM, FarmCPU, BLINK, etc.).

PLINK results: Association analysis results (*.assoc.linear).

rMVP results: GWAS results generated by rMVP. The p-value column is named after the trait by default and must be manually renamed to P.value.

Genotype files: SNP information, genotype matrix, and MAF files.

PCA files: Principal component coordinates and eigenvalues.

Annotation files: GTF annotation files for gene annotation.

LD decay files: LD decay statistics generated by PopLDdecay or compatible software.

(3) Output Files

Depending on the selected function, GPP generates publication-quality PDF figures, PNG images, interactive HTML plots, significant SNP tables, and gene annotation results.

6. Troubleshooting

File not found: Ensure that all required input files are located in the specified working directory.

Unsupported file format: Verify that input files follow the required format and contain the correct column names.

Missing annotation: Ensure that the GTF annotation file matches the reference genome used in the GWAS results.

Empty Manhattan plot: Check whether the p-value column is correct and whether the significance threshold is appropriate.

R version error: GPP requires R version 4.3.3 or higher.

Package dependency error: Install all required R packages before running GPP.

7. Explanation

GPP provides an integrated workflow for genomic data analysis and visualization. Different plotting functions automatically recognize the corresponding input file formats and generate publication-quality figures. Depending on the selected analysis, intermediate objects may be created during execution but are retained only in the R session unless explicitly exported.

For GWAS visualization, significant SNPs can be automatically identified according to the specified significance threshold, and nearby genes can be annotated using the supplied GTF file within a user-defined genomic window. For PCA analysis, GPP supports two-dimensional, three-dimensional, and interactive visualizations with customizable group colors and legends. For LD analysis, multiple LD decay result files can be plotted simultaneously for direct comparison among populations or datasets.

All generated figures are exported in high-resolution formats suitable for publication, and result tables are saved automatically when corresponding output parameters are specified.